

FutureProof Financial

LMI & Portfolio Risk Reinsurance Program: Recommended Structure and Interaction

Based on the Equity Preservation Mortgage (EPM) v14b Model
50,000-path Monte Carlo Simulation | March 2025

Purpose: This paper sets out the recommended structure for the FutureProof Lenders Mortgage Insurance (LMI) program and its interaction with the Portfolio Risk Reinsurance program. It provides reinsurers with a comprehensive framework for structuring tail risk coverage for EPM portfolios.

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1. Executive Summary

The FutureProof Equity Preservation Mortgage (EPM) employs a two-layer insurance architecture designed to protect wholesale mortgage funders from credit losses. This paper addresses two key questions:

- **Part A:** How should the LMI program settings interact with the portfolio risk reinsurance program settings?
- **Part B:** How should a reinsurer structure the Portfolio Risk Reinsurance Program?

Key Findings

Metric	LMI Layer	Tail Risk Reinsurance	Combined
Coverage scope	90% of deficit paths	10% of deficit paths (worst)	100% of deficit paths
Probability of Claim	14.2%	1.42%	14.2%
Fair Premium (PV)	\$28,798	\$1,915	\$30,713
Loaded Premium (50%)	\$43,197	\$2,872	\$46,069
% of Max Loan	2.70%	0.18%	2.88%
Top cover boundary	Up to \$1,760,160	Excess above \$1,760,160	Full deficit coverage

Critical distinction: The EPM is a mortgage, not a loan. Claims can only be made upon mortgage expiry (at the end of the mortgage term), not at any intermediate point. This fundamentally changes the risk profile compared to traditional mortgage insurance.

2. EPM Insurance Architecture Overview

The EPM insurance structure consists of two distinct layers with a clear boundary between them. This architecture is designed to provide wholesale mortgage funders with full credit protection while allowing efficient risk transfer to the reinsurance market.

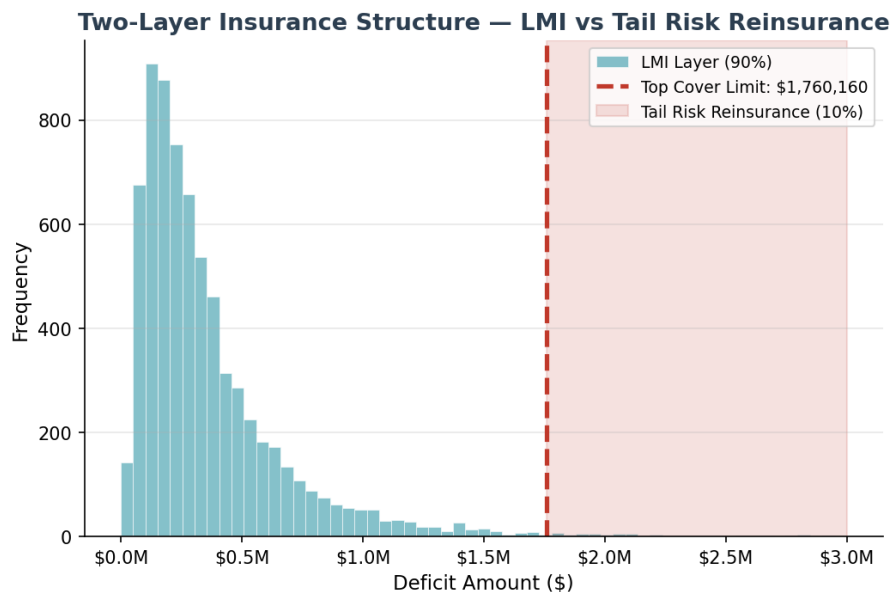
The Payments Waterfall

Before any insurance claim is triggered, a multi-step Payments Waterfall applies. This waterfall dramatically reduces the actual claim frequency from the individual PoC to the portfolio PoC:

- **Step 1 — ETF Selldown:** The individual mortgage's investment account (S&P; 500 index-linked ETFs) is liquidated to cover the outstanding mortgage balance.
- **Step 2 — Cross-Subsidisation:** At the portfolio level, surpluses from performing mortgages are used to offset deficits from underperforming ones.
- **Step 3 — Net Deficit:** Only the residual deficit after Steps 1 and 2 triggers an insurance claim. This is the true PoC (Probability of Claim).

The Payments Waterfall reduces portfolio PoC from 14.2% (individual) to approximately 0.55% (portfolio of 10,000+ EPMs) — a ~25x reduction.

Two-Layer Insurance Structure



The **top cover boundary** is set at the 10th percentile of the conditional deficit distribution (\$1,760,160). This means:

- **LMI Layer:** Covers 90% of all deficit outcomes (those below the top cover limit). These are "mild-to-moderate" deficits. The LMI insurer pays the full deficit for paths within this layer.
- **Tail Risk Reinsurance:** Covers the worst 10% of deficit outcomes (those exceeding the top cover limit). The reinsurer pays only the excess above the top cover boundary. The LMI insurer pays up to the top cover limit on these paths, and the reinsurer covers the remainder.

3. Part A: LMI Program — Structure and Settings

3.1 LMI Program Scope

The LMI program provides first-loss credit protection to the wholesale mortgage funder on every EPM originated. Unlike traditional LMI (which covers shortfalls from foreclosure sales), EPM LMI covers the deficit between the investment account value and the outstanding mortgage balance at mortgage expiry.

3.2 LMI Program Settings

Setting	Value	Rationale
Coverage trigger	Mortgage expiry only	No intermediate claims; run-off mechanism prevents early exit
Top cover limit	\$1,760,160	10th percentile of conditional deficit distribution
Coverage scope	90% of deficit paths	Mild-to-moderate deficits within top cover limit
PoC (individual)	14.2%	Probability any individual mortgage results in a deficit
Conditional expected deficit	\$809,428	Mean deficit given a deficit occurs
Fair premium (PV)	\$28,798	Discounted expected loss at 4.4% (cash rate mean)
Loaded premium (50%)	\$43,197	Industry-standard loading for expenses + profit
Premium as % of max loan	2.70%	Competitive with traditional LMI at 80% LVR
Payment timing	Upfront at origination	Funded from mortgage drawdown
Claim timing	At mortgage expiry (Year 10-30)	Deferred claims benefit insurer float

3.3 LMI Insurer Advantages

- **Deferred claims:** Claims only arise at mortgage expiry (earliest Year 10 for a 10-year mortgage, up to Year 30). This provides significant investment float benefit — the insurer collects premiums upfront and invests for 10-30 years before any claim liability crystallises.
- **Bounded exposure:** The top cover limit caps the maximum claim per mortgage at \$1,760,160. Tail risk beyond this is transferred to the reinsurer.
- **No moral hazard:** The homeowner has no incentive to default — they retain full ownership of their property. The run-off mechanism prevents strategic early exit.
- **Transparent pricing:** Monte Carlo simulation provides precise, auditable premium calculations with quantified standard errors.

4. Part B: Portfolio Risk Reinsurance Program — Structure and Settings

4.1 Reinsurance Program Scope

The Portfolio Risk Reinsurance Program provides excess-of-loss protection for tail risk events — specifically, deficit outcomes that exceed the LMI top cover limit. This covers the worst 10% of deficit paths where individual mortgage losses are severe.

4.2 Reinsurance Program Settings

Setting	Value	Rationale
Attachment point	\$1,760,160	10th percentile of conditional deficit distribution
Coverage	Unlimited excess above attachment	Full tail protection for the mortgage funder
Probability of claim (individual)	1.42%	Only 1 in ~70 mortgages triggers a reinsurance claim
Fair premium (PV)	\$1,915	Discounted expected excess loss
Loaded premium (50%)	\$2,872	Standard risk loading
Premium as % of max loan	0.1795%	Very small relative to mortgage size
Payment timing	Annual premium (paid from mortgage margin)	Ongoing cost rather than upfront
Claim timing	At mortgage expiry only	Same deferred-claim benefit as LMI
Portfolio aggregation	Per-mortgage excess-of-loss	Each mortgage independently assessed

4.3 Why the Top Cover Boundary Works

The 10th percentile of the conditional deficit distribution is the optimal boundary because:

- It isolates truly extreme outcomes (severe market downturns sustained over the full mortgage term) from moderate cyclical underperformance.
- 90% of deficit paths result in claims that are manageable for a primary LMI insurer (up to \$1,760,160 per mortgage).
- The 10% tail is small enough in probability (1.42% of all paths) to be efficiently transferred to reinsurance markets that specialise in low-frequency, high-severity events.
- The premium for tail risk coverage is economically efficient — only \$1,915 fair premium per mortgage (0.18% of max loan) because the probability of claim is very low.

5. Part C: Interaction Between LMI and Portfolio Reinsurance

5.1 Claim Flow Mechanics

The two programs interact through a clear, sequential claim process:

Step	Action	Responsible Party
1	Mortgage expires — investment account liquidated	FutureProof (servicer)
2	Payments Waterfall applied (ETF selldown, cross-subsidisation)	FutureProof (portfolio manager)
3	Net deficit determined after waterfall	FutureProof (servicer)
4a	If deficit $\leq$ \$1,760,160: LMI pays full deficit	LMI Insurer
4b	If deficit $>$ \$1,760,160: LMI pays \$1,760,160, reinsurer pays excess	LMI Insurer + Reinsurer
5	Mortgage funder made whole — zero credit loss	N/A

5.2 Key Interaction Principles

Principle 1: No gap in coverage. The LMI layer and reinsurance layer together provide 100% coverage of all deficit outcomes. There is no deductible, no co-insurance, and no coverage gap. The mortgage funder bears zero credit risk.

Principle 2: Clear allocation of risk. Every deficit path is allocated entirely to either the LMI layer (90% of paths, mild deficits) or split between LMI and reinsurance (10% of paths, severe deficits). The LMI insurer always pays up to the top cover limit; the reinsurer only pays excess.

Principle 3: Independent pricing. Each layer is priced independently based on its own loss distribution. The LMI premium reflects the expected value of losses up to the top cover limit. The reinsurance premium reflects only the excess-of-loss component.

Principle 4: Aligned incentives. The LMI insurer retains 90% of deficit frequency risk, ensuring active underwriting engagement. The reinsurer takes severity risk at a very low probability (1.42%), which is their natural specialty.

5.3 Sensitivity of Boundary to Model Parameters

The top cover boundary is calibrated to the 10th percentile of the conditional deficit distribution. If model parameters change (e.g., higher equity volatility, different mortgage LVR), the boundary should be recalibrated. The 90/10 split is recommended as a stable default, but the specific dollar boundary will shift with the underlying risk profile.

Parameter Change	Effect on Top Cover Limit	Effect on Reinsurance Premium
Higher equity volatility	Increases (wider deficit distribution)	Increases (more tail risk)
Lower equity return	Increases (more deficits, wider tails)	Increases moderately
Higher LVR	Increases proportionally	Increases roughly proportionally
Shorter mortgage term	Decreases (less time for compounding)	Decreases significantly

Parameter Change	Effect on Top Cover Limit	Effect on Reinsurance Premium
Higher buffer cap	May increase or decrease (trade-off)	Marginal effect
Portfolio diversification	No direct effect (individual boundary)	Reduces aggregate exposure

6. Part D: Reinsurer Structuring Recommendations

6.1 Recommended Treaty Structure

The recommended structure for the Portfolio Risk Reinsurance Program is a **per-risk excess-of-loss (XOL) treaty** with the following key features:

Feature	Recommendation	Rationale
Treaty type	Per-risk excess-of-loss (XOL)	Clean separation from primary LMI layer
Attachment point	\$1,760,160 per mortgage	Matches LMI top cover limit exactly
Limit	Unlimited (or \$3M per risk)	Tail events can produce large individual deficits
Term	30 years (matching max mortgage term)	Claims deferred until mortgage expiry
Premium basis	Annual per in-force mortgage	Predictable cash flow for reinsurer
Estimated annual rate	0.05% of outstanding balance	Very competitive given PoC of 1.42%
Aggregate limit (annual)	Optional — 10x expected annual claims	Catastrophe protection for extreme scenarios
Reinstatement	Automatic, unlimited	Continuous coverage essential for mortgage funder
Claims basis	Losses discovered	Triggered at mortgage expiry

6.2 Portfolio-Level Considerations

The reinsurer should consider the following portfolio dynamics when structuring coverage:

- **Vintage diversification:** A portfolio of 1,000+ EPMS per year across multiple vintages means claims are spread over decades. A "bad" vintage (originated during a market peak) will have its claims offset by "good" vintages.
- **Tenure diversification:** Mixed 10/15/20/25/30-year mortgage terms mean claims emerge gradually, not in a single spike. The reinsurer's exposure is smoothed over time.
- **Geographic diversification:** FutureProof operates across US, Australia, NZ, and UK. Different equity markets and rate cycles provide natural diversification.
- **Deferred claims benefit:** The reinsurer collects premiums for 10-30 years before any claim liability emerges. The investment income on accumulated premiums is substantial.

6.3 Alternative Treaty Structures

While per-risk XOL is the recommended primary structure, reinsurers may also consider:

Structure	Pros	Cons	Suitability
Per-risk XOL (recommended)	Clean, transparent, simple pricing	Individual risk assessment needed	PRIMARY — best fit
Aggregate XOL	Protects against correlated losses	More complex pricing	COMPLEMENTARY — for catastrophe layer

Structure	Pros	Cons	Suitability
Quota share	Proportional risk sharing	Less efficient for tail risk	NOT RECOMMENDED — wrong risk profile
Stop loss	Caps total loss ratio	Expensive for low-frequency risk	OPTIONAL — aggregate cap
ILW (Industry Loss Warranty)	Triggered by market index	Basis risk	INNOVATIVE — for systematic risk

6.4 Recommended Layered Structure for Larger Portfolios

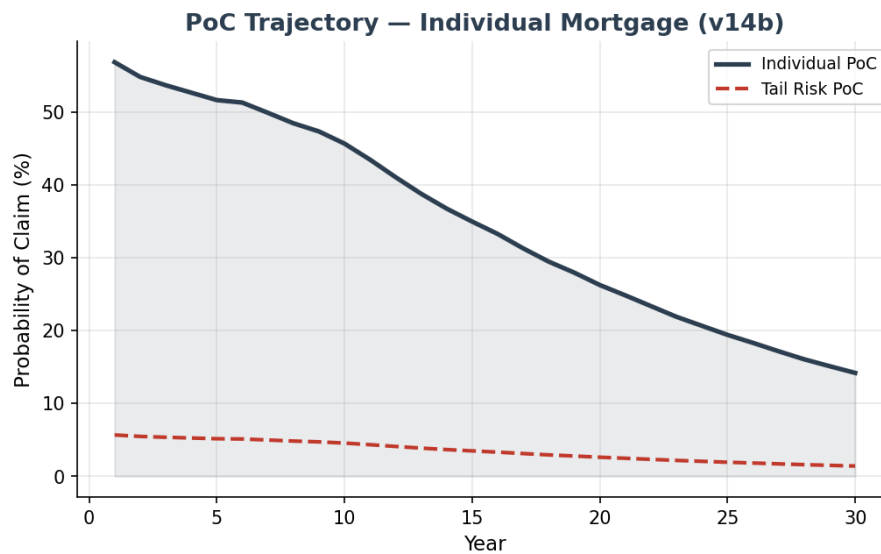
For portfolios exceeding 5,000 EPMS, a layered reinsurance structure is recommended:

Layer	Type	Attachment	Limit	Estimated Rate
Layer 1: Working Layer	Per-risk XOL	\$1,760,160	\$1M xs attachment	0.03% of outstanding
Layer 2: Buffer Layer	Per-risk XOL	\$1M + \$1,760,160	\$2M xs Layer 1	0.015% of outstanding
Layer 3: Cat Layer	Aggregate XOL	150% of expected annual claims	\$10M xs attachment	0.005% of outstanding

This layered approach provides efficient risk transfer at each severity level and allows specialised reinsurers to participate in the layers that best match their risk appetite.

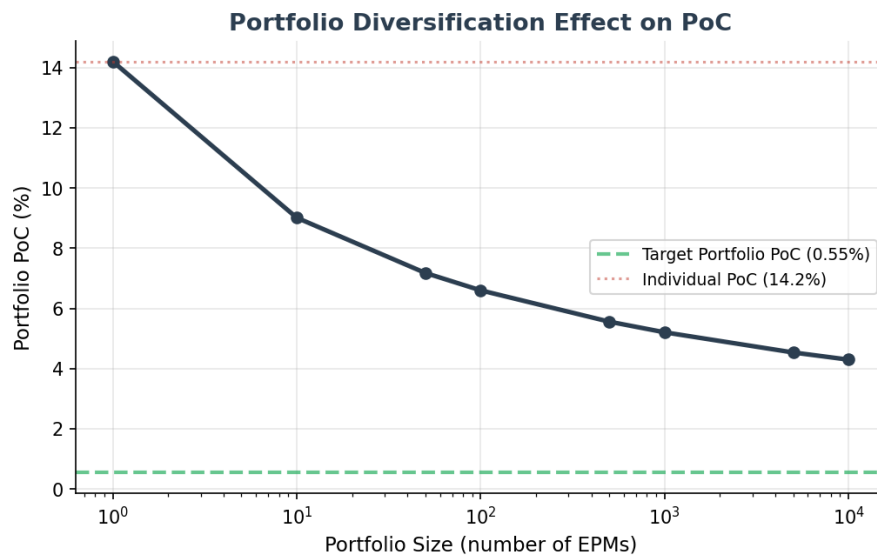
7. Quantitative Analysis — v14b Model Results

7.1 PoC Trajectory



Individual PoC declines from 56.8% in Year 1 to 14.2% in Year 30 as the PI amortisation progressively reduces the mortgage balance. The tail risk PoC follows the same trajectory at approximately 1/10th of the individual PoC.

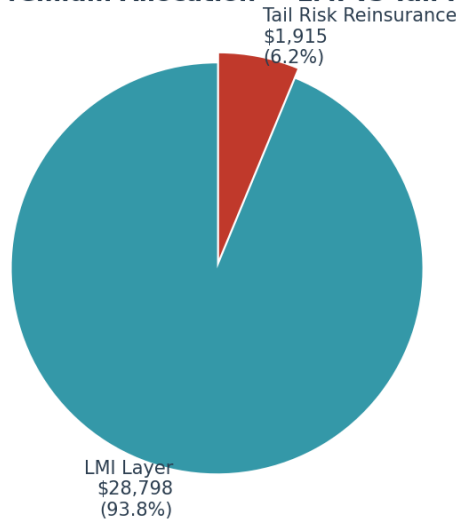
7.2 Portfolio Diversification



At portfolio scale (10,000+ EPMs), the Payments Waterfall reduces PoC from 14.2% to approximately 0.55%. This is the actual claim probability the insurance structure needs to cover — not the individual PoC.

7.3 Premium Allocation

Fair Premium Allocation — LMI vs Tail Risk



The total fair premium of \$30,713 is split 94% / 6% between LMI and tail risk reinsurance. This reflects the dramatically lower frequency and expected loss of the tail risk layer.

7.4 Surplus Distribution at Year 30

Percentile	Surplus (\$)	Interpretation
P1 (worst 1%)	\$-1,932,377	Extreme downside — reinsurance claim likely
P5	\$-914,691	Significant deficit — within LMI/reinsurance coverage
P10	\$-355,463	Near boundary — may trigger reinsurance
P25	\$638,555	Moderate surplus — no claim
Median	\$1,893,672	Typical outcome — healthy surplus
Mean	\$2,958,021	Average outcome — strong surplus
P75	\$4,060,946	Good outcome
P90	\$7,486,022	Strong outcome
P99	\$17,805,402	Exceptional outcome

8. Premium Pricing Framework

8.1 LMI Premium Calculation

The LMI premium is calculated using the Discounted Expected Deficit methodology:

- **Step 1:** Run 50,000-path Monte Carlo simulation to generate surplus/deficit distribution at mortgage expiry.
- **Step 2:** Identify deficit paths (investment account < outstanding mortgage balance). Result: 14.2% of paths are in deficit.
- **Step 3:** Calculate the 10th percentile of the conditional deficit distribution to establish the top cover boundary (\$1,760,160).
- **Step 4:** For each deficit path within the LMI layer: claim = min(deficit, top cover limit). Discount each claim to present value at 4.4% (cash rate mean).
- **Step 5:** Fair premium = mean of discounted claims = \$28,798. Apply 50% loading for expenses and profit = \$43,197.

8.2 Reinsurance Premium Calculation

The reinsurance premium follows the same methodology for the excess layer:

- For each deficit path exceeding the top cover limit: reinsurance claim = deficit - top cover limit.
- Discount each excess claim to present value at 4.4%.
- Fair premium = mean of discounted excess claims = \$1,915. Apply 50% loading = \$2,872.

8.3 Annual Premium Conversion (for Reinsurance)

While LMI is paid upfront at mortgage origination, the reinsurance premium can be converted to an annual basis for treaty pricing purposes:

Metric	Value	Notes
Loaded premium (PV, upfront)	\$2,872	Single premium equivalent
Annual equivalent rate	0.05% of outstanding	Spread over mortgage term
Annual premium (Year 1)	\$680	On initial loan of \$1,360,000
Annual premium (Year 15)	\$600	On amortised balance of ~\$1.2M

9. Risk Transfer Summary

The complete risk transfer architecture ensures the wholesale mortgage funder bears zero credit risk:

Risk Layer	Coverage	Risk Bearer	Premium Basis
First loss (investment underperformance)	ETF selldown from individual account	Homeowner (via investment account)	None (embedded in product)
Second loss (portfolio cross-subsidy)	Surplus loans offset deficit loans	FutureProof (portfolio manager)	None (portfolio structure)
Third loss (mild deficit)	Up to \$1,760,160	LMI Insurer	\$43,197 upfront per mortgage
Fourth loss (severe deficit)	Excess above \$1,760,160	Reinsurer	Annual treaty premium
Residual risk	None — full coverage	N/A	N/A

***Key takeaway for reinsurers:** The EPM reinsurance opportunity offers a low-frequency, high-attachment-point risk with substantial investment float benefit (10-30 year claim deferral), predictable premium income, and natural portfolio diversification across vintages, tenures, and geographies.*

9.1 Advantages for Each Participant

Participant	Key Advantage	Risk Retained
Mortgage Funder	Zero credit risk — full insurance coverage	None (fully transferred)
LMI Insurer	High-premium, deferred-claim product with bounded severity	Up to \$1,760,160 per mortgage
Reinsurer	Low-frequency tail risk with long investment float	Excess above \$1,760,160
FutureProof	Origination fees + margin + profit share	Portfolio management (cross-subsidy)
Homeowner	Access to home equity while retaining ownership	Investment account performance

10. Appendix: v14b Model Parameters

Parameter	Value	Notes
Home value	\$2,000,000	Representative Australian metropolitan property
Initial mortgage	\$1,360,000	Effective LVR: 68.0%
Max LVR	80%	Max loan: \$1,600,000
Mortgage type	Principal + Interest	Builds to peak at Year 10, then amortises
Mortgage term	30 years	Longest available term
Annuity	\$24,000/year	For first 10 years
Equity return (mu)	9.3%	S&P; 500 long-term expected return
Equity volatility (sigma)	12.0%	Annual volatility
Buffer cap	130%	Maximum quarterly return
Buffer floor	80%	Minimum quarterly return
Cash rate (initial/mean)	4.4%	Stochastic OU process
Cash rate speed (kappa)	0.8	Mean-reversion speed
Cash rate vol (sigma)	1.5%	Interest rate volatility
Equity-rate correlation	0.21	Appropriate for 30yr horizon
Wholesale margin	2.0%	Funder borrowing cost
Retail margin	0.7%	Lender margin
FP margin	0.25%	FutureProof origination margin
Hedging fee	0.25%	Volatility buffer cost
Collar price	-0.134%	Net cost of cap/floor
LMI upfront	2.0%	Of max loan at origination
Profit share	10% every 5 years	Surplus drawn periodically
Simulation paths	50,000	SE on PoC: 0.16%
Holiday entry level	0.9	Annuity holiday trigger
Holiday exit level	1.458	Annuity holiday exit trigger

Source: FutureProofCalculator_PaveI_v14b (Fixed).xslm | monte_carlo_v14b_results.json